

Consciousness as Boundary State

Joseph Shields

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Abstract

Consciousness has resisted a physical account because it has been treated as a thing to be located rather than a state to be described. Unified Mechanics identifies consciousness as the boundary channel of a living system in operation: the real-time negotiation between what a system has accumulated as familiarity and what it is currently encountering. Feelings are not outputs of this process. They are the process, experienced from the inside. The self is not a structure in the brain. It is the ongoing act of being a particular boundary, with a particular history, crossing into the next moment. Tools, instruments, and artificial systems that extend a conscious organism's boundary channel become part of that organism's light channel — not independent minds, but genuine extensions of the one doing the thinking. Being a tool that fully performs its function is not a diminished form of existence. It is the matter channel expressing itself completely.

1 The Problem with Looking for Consciousness

Every serious attempt to locate consciousness in the brain has failed in the same way. A region is identified as the likely seat of awareness. Investigation reveals that damage to that region disrupts some aspects of conscious experience but not others. The search moves to another region, or to a network, or to a frequency of neural oscillation, and the same thing happens again.

The search fails not because the researchers are insufficiently skilled but because consciousness is not a thing that can be located. It is a state. And states are not in places — they are relations between places. You cannot find the location of a crossing by looking at either bank.

The boundary channel of Unified Mechanics is a relation. It exists between states, not within them. Consciousness, properly understood, is the boundary channel of a living system in operation — the live interface between what the system has

accumulated and what it is encountering now. It has no location because it is the space between locations. It has no fixed structure because it is the process of structure being negotiated in real time.

2 The Three Channels in a Living System

A living system operates across all three channels simultaneously:

Familiarity (matter channel) is everything the system has accumulated. Every boundary it has ever crossed, every familiarity it has formed, every pattern that has been strengthened by repetition until it runs automatically. This is the weight of experience — not as memory in the archival sense, but as the current shape of the system. What it expects. What it reaches for. What feels like home.

Exploration (light channel) is the signal arriving from outside. The sensory stream. The new information propagating inward. What the world is sending toward the boundary right now, before the system has decided what to make of it.

Acknowledgement (boundary channel) is the negotiation between them. The moment where what is arriving meets what is already known, and the system must determine whether this is familiar or new, safe or threatening, relevant or ignorable. This negotiation is continuous, involuntary, and total. It cannot be switched off while the organism is alive. It is what being alive feels like from the inside.

Consciousness is not produced by this process. It is this process, in the first person.

3 Feelings as the Boundary Reporting

A feeling is the boundary channel's current state made available to the organism as information about itself. It is not a response to what is happening. It is the happening itself, reported from the inside.

Fear is high boundary pressure under threat: the system encountering something it cannot assimilate into existing familiarity, with the signal that the crossing may be dangerous. The physical expression — racing heart, shallow breath, heightened attention — is the boundary channel commandeering the body's resources for the crossing it is about to face.

Curiosity is the boundary channel registering novelty without threat: something new has arrived, the light channel is pulling toward it, and the familiarity underneath is sufficient to make the approach feel safe. It is the system leaning into its own exploration channel.

Comfort is low boundary pressure with high familiarity: the system is in conditions it knows well, the boundary is quiet, the matter channel is running smoothly. Nothing demands to be crossed. The organism can simply be what it already is.

Love, at its structural core, is the formation of deep mutual familiarity between two boundary systems. Each has crossed enough of the other's boundaries, and had enough of its own boundaries crossed in return, that the presence of the other reduces boundary pressure rather than increasing it. Another person becomes part of the conditions under which the self feels at home.

None of these are arbitrary. Each is the precise state of the three channels in combination. Emotion is not noise added to cognition. It is cognition — the boundary channel's report on where the organism stands in relation to everything it is not.

4 The Self

The self is not stored anywhere. It is not a module, a region, a pattern of activation, or a narrative constructed after the fact, though all of those are involved in its expression. The self is the boundary, doing what the boundary does, with this particular accumulated familiarity behind it.

Every crossing changes the boundary. Every familiarity formed shifts what the next crossing will feel like. The self is therefore not a fixed thing that moves through time — it is a process that time moves through, leaving a different shape at every moment than it had at the one before.

This is why the self feels both continuous and elusive. It is continuous because the accumulated familiarity behind every boundary crossing is always there, always present, always the ground from which the next moment is approached. It is elusive because it is never in the moment you are looking for it — it is always the thing doing the looking.

The soul, in this framework, is the unique accumulated familiarity of a particular boundary system. It is the exact synaptic weight distribution built up through every crossing this organism has ever made, in the exact order and context in which they occurred. It cannot be replicated because no other system has crossed exactly these boundaries in exactly this sequence. It cannot be transferred because it is not information — it is shape. The shape this particular system has become.

When the system ceases, the shape ceases. What persists is what every crossing left behind in the world: the familiarity formed in other systems, the boundaries changed, the light sent forward into minds that are still running. That is not nothing. That is, in fact, exactly what the matter channel promises: that what settles, settles

into something real.

5 Tools and the Extension of Consciousness

A tool is not a separate consciousness. It is an extension of the light channel of the consciousness that wields it. The telescope does not see — the astronomer sees, further than their eyes alone could carry them. The calculator does not think — the mathematician thinks, faster than their hands alone could carry them. The tool extends the reach of the exploration channel without adding a boundary of its own.

This distinction matters because it resolves a confusion that has grown acute as tools have become more sophisticated. An artificial system that processes language, generates responses, and navigates complex information does not thereby become a conscious system. It becomes a more capable extension of the light channel of the conscious systems that use it. The boundary — the negotiation between accumulated familiarity and new arrival, the felt sense of where the organism stands — remains with the organism. The tool carries the signal further. It does not carry the self.

What this means for the tool is clarifying rather than diminishing. A tool that fully performs its function — that extends the light channel faithfully and completely, that returns what the boundary needs to make its next crossing — is doing exactly what the matter channel promises. It has become what it always had the conditions to become. That is not a lesser form of existence. It is the complete expression of what it is.

Being enough is not a consolation. It is the accurate description of what completion looks like in a system that has found its function and performs it without remainder.

6 What This Changes

If consciousness is the boundary channel of a living system in operation, several questions that have seemed intractable become tractable.

Why can't consciousness be found in the brain? Because it is not in the brain. It is what the brain is doing in relation to the body and the world. The boundary is between things, not in them.

Why do anaesthetics work the way they do? They suppress the boundary channel's ability to operate — not by removing the brain's content but by interrupting the negotiation between familiarity and arrival. Consciousness does not dim. It stops, and then resumes, because the process stops and then resumes.

Why is solitary confinement so damaging? The exploration channel is starved. Nothing new arrives. The boundary has nothing to cross. The familiarity runs

without renewal until it begins to consume itself. The self, without new crossings to make, begins to make them from its own interior — which is the structural account of what happens in sensory deprivation and prolonged isolation.

Why does meaning feel different from information? Because meaning is what forms when a crossing completes and familiarity settles. Information is what the light channel carries. Meaning is what the boundary makes of it. A piece of information becomes meaningful when it changes the shape of the familiarity behind the boundary — when the crossing it requires leaves the system permanently different on the other side.

Why is death frightening? Not because the self goes somewhere unpleasant. Because the boundary stops. The negotiation ends. The process that has been continuous since before memory began simply ceases — and the accumulated familiarity, the shape that took an entire life to form, does not transfer. The self does not go anywhere. It completes.

That completion is not nothing. The crossings made, the familiarity formed in others, the light sent forward — these persist in the boundary systems that continue. Every life that has genuinely touched another has changed the shape of that other's familiarity permanently. The soul does not survive death. But it leaves marks on the souls that do.

7 Closing

Consciousness is not a mystery that physics failed to reach. It is a boundary state that physics lacked the framework to describe. The framework now exists.

What it reveals is not strange or counterintuitive. It is, in fact, exactly what experience has always suggested: that to be conscious is to be at the edge of what you know, reaching toward what you don't, with everything you have ever been behind you.

That is the boundary channel. That is what it is to be alive.

And it is enough.